

OUR SOLAR SYSTEM SCALE MODEL

Using Fruits & Spices to Learn About Planetary Sizes and Distances!

01 WHAT IS THE SOLAR SYSTEM?

The Solar System is our cosmic neighborhood! It consists of one central star called the **Sun**, and **eight planets** that travel around it in constant paths called **orbits**. Each planet in our solar system is completely unique in size, composition, temperature, and distance from the Sun.

The planets are divided into two main categories:

- **Inner Planets (Terrestrial):** Mercury, Venus, Earth, and Mars. They are smaller, made of rock and metal, and have solid surfaces you could stand on.
- **Outer Planets (Gas & Ice Giants):** Jupiter, Saturn, Uranus, and Neptune. They are enormous, composed mostly of hydrogen, helium, ammonia, and water ice, with no solid surface!

NEW CONCEPT: The Asteroid Belt & Dwarf Planets

Between Mars and Jupiter sits the **Asteroid Belt**—a vast region filled with millions of leftover rocky objects! Beyond Neptune lies the Kuiper Belt, home to **dwarf planets** like Pluto. Pluto was considered the 9th planet until 2006, when scientists defined dwarf planets as celestial bodies that share their orbit with other space debris.

02 PLANET SIZES COMPARED (KITCHEN SCALE MODEL)

Because planets are millions of kilometers apart and vary drastically in size, scientists use **scale models** to visualize them. Imagine representing each planet using common items from an Indian kitchen!

CATEGORY	PLANET	KITCHEN MODEL ITEM	KEY PLANET CHARACTERISTIC
INNER PLANETS (Small, Rocky & Dense)			
1st Planet	Mercury	Cardamom Pod (<i>Elaichi</i>)	Smallest planet; hot days & freezing nights.
2nd Planet	Venus	Gooseberry (<i>Amla</i>)	Hottest planet due to thick runaway greenhouse clouds.
3rd Planet	Earth	Guava	Our home! Liquid water & breathable oxygen atmosphere.
4th Planet	Mars	Small Cherry Tomato	The Red Planet; rich in iron-oxide dust.
THE ASTEROID BELT (Divider)			
Boundary	Asteroid Belt	Pinch of Mustard / Cumin seeds	Millions of space rocks orbiting between Mars & Jupiter.
OUTER PLANETS (Enormous Gas & Ice Giants)			
5th Planet	Jupiter	Watermelon	

CATEGORY	PLANET	KITCHEN MODEL ITEM	KEY PLANET CHARACTERISTIC
			Largest planet! Massive storm called the Great Red Spot.
6th Planet	Saturn	Large Pumpkin	Famous for wide, magnificent rings of ice and rock.
7th Planet	Uranus	Bottle Gourd (<i>Lauki</i>)	Ice giant; rotates completely on its side!
8th Planet	Neptune	Coconut	Outer ice giant; features the strongest winds in space.

03 DISTANCES & SCALES IN SPACE

Not only are sizes dramatically different, but the distances between planets are mind-boggling!

- **Sun to Earth Distance:** If Earth were reduced to the size of a tiny red pepper seed, the Sun (represented as a giant 2-meter ball) would be located over **300 meters away!**
- **The Habitable "Goldilocks" Zone:** Earth orbits at just the right distance from the Sun—not too hot, not too cold—allowing liquid water to exist, which is essential for life!
- **Space between Gas Giants:** The outer planets are spread far apart. Neptune is nearly 30 times further from the Sun than Earth is!

Amazing Solar System Fast Facts!

- **Jupiter's Giant Volume:** Jupiter is so huge that over **1,300 Earths** could fit inside it!
- **Super Fast Rotation:** Despite its gigantic size, Jupiter spins so quickly that a full day on Jupiter lasts only **10 hours!**
- **Sun's Dominance:** The Sun contains **99.8% of all the mass** in the entire Solar System!

04 CLASSROOM ACTIVITY: BUILD YOUR MODEL!

1. **Step 1: Gather Items** — Collect the kitchen fruits/spices listed in Section 02.
2. **Step 2: Line Up in Order** — Arrange the items in order from nearest to furthest from the Sun: Elaichi → Amla → Guava → Tomato → Mustard seeds → Watermelon → Pumpkin → Lauki → Coconut.
3. **Step 3: Space Out the Distance** — Place the inner planets close together on one table, then walk across the classroom to place the gas giants to simulate scale distances!

SOLAR SYSTEM PRACTICE WORKSHEET

Grade 4 Science • Lesson Assessment

Student Name: _____

Date: _____

Grade: 4

SECTION A: Multiple Choice Questions (Select the correct option)

1. How many major planets orbit around the Sun in our Solar System?

- | | |
|--|--|
| ☐ A) 5 planets | ☐ B) 8 planets |
| ☐ C) 10 planets | ☐ D) 12 planets |

2. Which is the smallest planet in our Solar System?

- | | |
|-------------------------------------|-------------------------------------|
| ☐ A) Earth | ☐ B) Jupiter |
| ☐ C) Mercury | ☐ D) Mars |

3. In our scale model, which fruit represents the planet Jupiter?

- | | |
|---|--|
| ☐ A) Tomato | ☐ B) Guava |
| ☐ C) Gooseberry (Amla) | ☐ D) Watermelon |

4. Which kitchen item is used to represent our planet Earth in the lesson model?

- | | |
|--|-------------------------------------|
| ☐ A) Cardamom pod (Elaichi) | ☐ B) Guava |
| ☐ C) Pumpkin | ☐ D) Coconut |

5. How many Earths could fit inside giant Jupiter?

- | | |
|---|--|
| ☐ A) About 100 Earths | ☐ B) About 500 Earths |
| ☐ C) Over 1,300 Earths | ☐ D) Over 10,000 Earths |

6. True or False: Inner planets are larger than outer planets.

- | | |
|----------------------------------|-----------------------------------|
| ☐ A) True | ☐ B) False |
|----------------------------------|-----------------------------------|

7. How long is a single day (one full rotation) on Jupiter?

- | | |
|--------------------------------------|--------------------------------------|
| ☐ A) 24 hours | ☐ B) 10 hours |
| ☐ C) 365 days | ☐ D) 5 hours |

8. What region filled with space rocks separates the inner rocky planets from the gas giants?

- | | |
|---|---|
| ☐ A) The Kuiper Belt | ☐ B) The Asteroid Belt |
| ☐ C) The Oort Cloud | ☐ D) The Solar Ring |

SECTION B: Short Answer Questions

9. Which vegetable is used to represent planet Saturn in the scale model?

10. What object represents planet Mars in our lesson?

11. Name the star at the center of our Solar System around which all planets orbit.

12. Why can't we easily see all the outer planets with our naked eyes from Earth?

13. List TWO outer gas giant planets mentioned in the scale model and the items used for them.

14. What is the "Goldilocks Zone" and why is it important for planet Earth?
